

# Win back up to **R\$ 0.8 M a year** in sales that vanish from the shelf today

The Replenishment Center warns **2.9 days ahead** that an item will run out. The cheapest fix for each case comes ready, and the replenishment manager approves it **in one click**.

## TODAY

**4.1%**

of sales lost to stockouts: R\$ 2.39 M/yr on the top-selling items

## CHAIN-WIDE

**~R\$ 1.4 M**

recoverable per R\$ 100 M of sales ( $\frac{1}{3}$  of the loss)

## THE ASK

**30 days**

pilot in 4 stores, with 4 control stores

**Where the numbers come from:** a prototype already running on Databricks, on the **last 120 days of history from all 20 stores**, for the 129 top-selling items. The pilot measures the result in store operations.

# The answer on one page

## THE PROBLEM

Across the 129 top-selling items in all 20 stores, **4.1% of sales is lost to stockouts: R\$ 2.39 M a year**, in line with retail's ~4%. The team finds out **after** the shelf is empty.

## THE VALUE

Up to **⅓ of the loss recovered**: R\$ 0.4 to 0.8 M a year on those items, depending on execution; **R\$ 0.7 to 1.4 M per R\$ 100 M of sales**. Measured platform cost: **~US\$ 440 a month**, at list price.

## THE SOLUTION

A model flags **which in-stock items will run out in the next 7 days**. An AI agent checks the facts and proposes the cheapest fix: **transfer, expedite or urgent order**. The manager approves in one click. **All on a single platform, Databricks.**

## THE ASK

**A 30-day pilot in 4 stores**, with 4 control stores. The decision to scale comes from LojaBR's own numbers, not from a projection.

# An empty shelf is a sale that goes to the competitor

SALES LOST

4.1%

of sales, to stockouts

PER YEAR

R\$ 2.39 M

20 stores × 129 items

AVAILABILITY · 7 DAYS

93.5%

6.5% stockout rate (store × item × day)

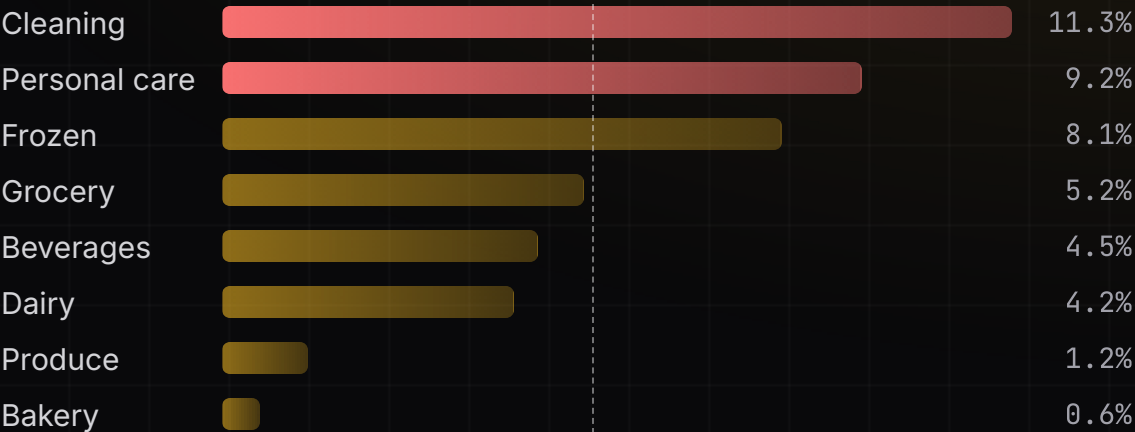
WORST SUPPLIER

63%

of deliveries on time

120 days of sales, stock and purchase-order history from all 20 stores, for the 129 top-selling items.

STOCKOUT RATE BY CATEGORY · 120 DAYS · CHAIN AVERAGE 5.3%



**The loss is concentrated:** cleaning and personal care run out ~2× the average. Their suppliers deliver on time only **63%** and **68%** of the time.

# What moves in the numbers LojaBR already tracks

| KPI                                | TODAY                         | WITH THE CENTER · 50% TO 100% EXECUTED    | WHO TRACKS IT         |
|------------------------------------|-------------------------------|-------------------------------------------|-----------------------|
| On-shelf availability              | 93.5%                         | 94.7% to 95.8%                            | VP of Operations      |
| Stockout rate (store × item × day) | 6.5%                          | 5.3% to 4.2%                              | VP and manager        |
| Sales lost to stockouts            | 4.1% · R\$ 2.39 M/yr          | -18% to -35% · R\$ 0.42 to 0.84 M/yr back | VP and finance        |
| Warning before the shelf empties   | none (reactive)               | ~2.9 days · 75% ≥ 2 days                  | Replenishment manager |
| Items to review per day            | 29% of assortment · 38% right | 8% · 84% right                            | Replenishment manager |

Ranges go from 50% to 100% of actions executed on time (slide 9). Assumption: stockouts fall in the same proportion as recovered sales. Rule of thumb: every **R\$ 100 M of sales** loses ~R\$ 4 M to stockouts; recovering a third is worth **~R\$ 1.4 M**.

# What changes for each of them

## EXECUTIVE SPONSOR

### VP of Operations & Supply Chain

- **Sales back on the P&L:** R\$ 0.42 to 0.84 M/yr on the top-selling items, ~R\$ 1.4 M per R\$ 100 M of sales.
- **The value comes cheap:** 80% of it comes from expediting orders already on their way, with no new purchase and no freight.
- **Risk under control:** the manager decides, and every decision is logged with author and time.
- **Measured cost:** ~US\$ 440 a month to run everything, at list price.

## DOMAIN OWNER

### Replenishment Manager

- **A short list:** 8% of items, 3.6× fewer than today's rule, sorted by R\$ or by urgency.
- **2.9 days of warning:** time to transfer or expedite, instead of firefighting.
- **The reasoning, ready:** stock, donor store and supplier punctuality on the card itself.
- **One click approves,** and data questions go to Genie in plain Portuguese, no ticket needed.

# An action queue, not a report

- 1** **Opens the queue at 7 am.** All of the day's actions, sorted by R\$ protected or by what empties first.
- 2** **Reads the card.** Product, store, the cheapest fix, the agent's reasoning and the R\$ at stake.
- 3** **Approves in one click.** The decision is logged with their name, and **"R\$ protected today"** goes up on screen.

## Transfer

from a nearby store  
with excess

## Expedite

the order already on its  
way

## Urgent order

when there is no  
alternative

CRÍTICO

LIMPEZA

#3

↔ Transferir entre lojas

**Sabão em Pó Omo Lavagem Perfeita 1,6kg**

Barra da Tijuca (16,3 d)

 → 

Tatuapé (0,2 d)

 333 km

risco 7 dias  
96%

mover  
**18 un**

chegada  
2 dias

estoque 1 un · 46 un a caminho · chega em ~4 d · pontualidade do fornecedor 64%

🚀 AGENTE DE REPOSIÇÃO · VERIFICOU 3 FONTES

A loja Tatuapé está com apenas 1 unidade em estoque (cobertura de 0,2 dia) e o pedido a caminho só chega em 2 dias, com fornecedor Limpa Mais tendo apenas 63,6% de pontualidade e atraso médio de 4 dias. A LojaBR Barra da Tijuca tem excedente de 39 unidades e cobertura de 16,3 dias, permitindo transferir 18 unidades sem comprometer seu próprio estoque.

PROTEGE · PRÓXIMOS 7 DIAS

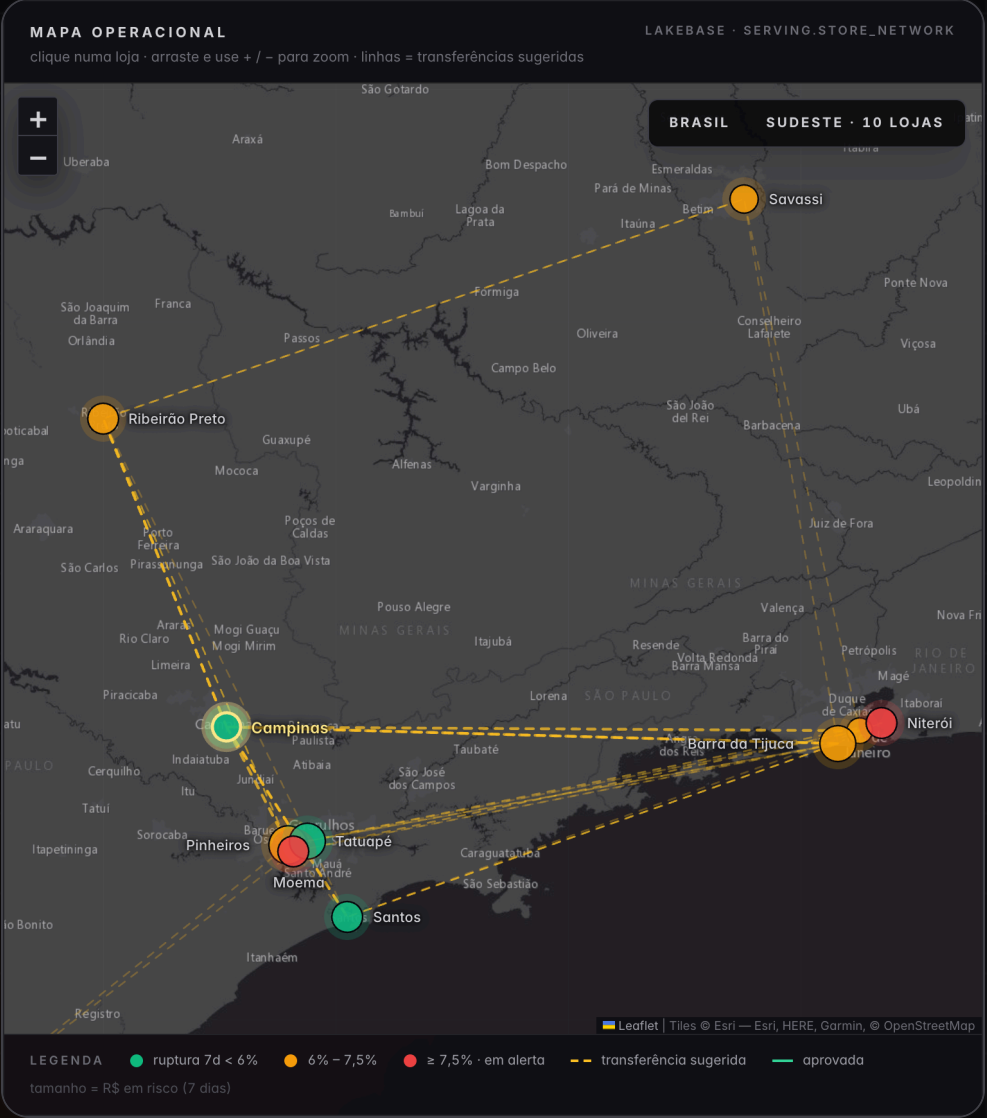
**R\$ 516**

cobre todo o risco previsto do item

×

✓ APROVAR

WHERE TO ACT FIRST



APP SCREEN · SOUTHEAST ZOOM

# Idle stock in one store fixes the gap in another

- **39 transfers** suggested in a single day, using other stores' excess.
- Only within **450 km**, never leaving the donor with less than **10 days** of stock, and arriving in **1 to 2 days**.
- The map shows stockouts per store and the routes. One click opens the store and its actions.

WHEN A TRANSFER PAYS OFF

Each transfer protects **R\$ 74** on average. It only pays off if the marginal freight costs less than that. So in the pilot we only transfer on routes that already run, such as the DC delivery, or by grouping the items of the same store pair.

Transfers are 18% of the value. Most of it (80%) comes from expediting orders already on their way.

## WHY TRUST THE ALERT

# The model is right more than twice as often as today's rule

### PRECISION · DAILY LIST

84%

vs 38% for the rule "cover < lead time"

### WARNING LEAD

2.9 days

3,691 correct alerts in the test; 75%  $\geq 2$  days ahead

### AUC

0.85

validated on 49,733 cases from 4 weeks the model never saw

### LIST SIZE

8%

of in-stock items, vs 29% for the rule

### AND WHEN IT IS WRONG?

A wrong alert does not become a loss. The donor keeps at least 10 days of stock, and the transferred item sells at the destination store. The worst case is the freight, hence the route rule on slide 7.

### HOW IT STAYS GOOD

The model is **retrained every day** in the same job, with a new version registered in Unity Catalog. The agent looks up stock, nearby stores and the supplier before recommending, and the system rejects any decision made before that lookup.

# What it is worth, by execution, and what it costs

| ACTIONS APPROVED AND EXECUTED ON TIME | 50 %                                | 75 %       | 100 %      |
|---------------------------------------|-------------------------------------|------------|------------|
| Sales protected per week              | R\$ 8.1 k                           | R\$ 12.1 k | R\$ 16.2 k |
| Per year · 20 stores × 129 items      | R\$ 0.42 M                          | R\$ 0.63 M | R\$ 0.84 M |
| Share of the annual loss recovered    | 18%                                 | 26%        | 35%        |
| Per R\$ 100 M of chain sales          | R\$ 0.7 M                           | R\$ 1.1 M  | R\$ 1.4 M  |
| Measured platform cost (list price)   | ~US\$ 440 a month in every scenario |            |            |

WHERE ONE DAY'S QUEUE GETS ITS VALUE



142 expedites    39 transfers (R\$ 2.9 k)    7 urgent orders (R\$ 0.4 k)

**Assumptions:** **expected** value (risk-weighted), not guaranteed; based on the queue of the 2026-09-28 run (R\$ 16.2 k over 7 days, ≈ 27% of the previous week's loss); top-selling items only; transfer freight left out, hence the route rule.

# From raw data to a decision, in one daily job



ARCHITECTURE VIEW IN THE APP (LABELS IN PORTUGUESE) · EVERY STAGE OPENS IN THE WORKSPACE

## ONE JOB, EVERY DAY AT 06:00

With data through yesterday, the queue is ready before stores open. Ingestion, quality, model, agent and publishing run together in 8.4 min.

## GOVERNED IN UNITY CATALOG

Lineage for every number, 7 data-quality rules (0 failures in the last run) and least-privilege access for the app.

## NO COPIES, NO SPREADSHEETS

The same data feeds the app, Genie and the model. Every stage opens in the workspace straight from the app.

**Every number in this deck is traceable:** it comes from the 2026-09-28 evidence run and the workspace's billing tables. The executed notebooks, with their outputs, and the cost queries are in the repository as text (evidence/).

# What the platform changes in the outcome

|               | WITH SEPARATE TOOLS                                                                                   | WITH DATABRICKS                                                                                        |
|---------------|-------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------|
| Integration   | ETL, data warehouse, ML, operational database, LLM, BI and app hosting: 7 pieces to integrate and run | One platform: Lakeflow, Unity Catalog, MLflow, Foundation Model API, Lakebase, Genie and Apps          |
| The number    | Copies between systems; the R\$ in the app does not always match the report                           | One governed dataset: the app, Genie and the model read the same number, with lineage                  |
| Time to pilot | Months of integration and one security review per tool                                                | The prototype already runs end to end; the pilot takes 30 days                                         |
| Security      | Permissions and audit spread across every tool                                                        | One place: who sees what, and who approved each action                                                 |
| Cost          | Fixed licences and infrastructure, running or not                                                     | Measured: <b>~US\$ 440 a month</b> at list price; the daily batch journey costs <b>US\$ 0.52 a day</b> |
| Exit          | Closed formats; migrating is a project                                                                | Open formats: Delta tables (and Iceberg via UniForm), standard Postgres, MLflow and code in Git        |

The same data already serves the next use cases:

demand forecasting

markdown of excess stock

store-level assortment

# Controlled risk, measured result

| WHEN      | WHAT HAPPENS                                                                           | HOW WE MEASURE                             |
|-----------|----------------------------------------------------------------------------------------|--------------------------------------------|
| Week 1    | Connect sales, stock and orders for the 4 pilot stores; wire the approval into the ERP | data arriving daily, quality $\geq$ 99%    |
| Weeks 2–4 | Action queue in the 4 stores; 4 similar stores as <b>control</b>                       | stockouts and lost sales, pilot vs control |
| End       | Decision to scale                                                                      | R\$ recovered per store, against cost      |

SUCCESS CRITERION

Cumulative stockout rate over weeks 2–4  $\geq$  **1.5 p.p. below** control, and R\$ recovered greater than the cost.

CHOOSING THE STORES

4 stores with above-average stockouts, across 2 regions and 2 formats. Control is matched on format, region and stockout rate and lives the same month, so seasonality and promotions cancel out.

RISKS AND MITIGATION

Model is wrong: the manager decides. Adoption: one click, in the morning routine. Freight: existing routes only. Incomplete supplier data: the rule covers it.

## THE ASK

# Sponsor a **30-day pilot in 4 stores**, with a control group.

Measure stockouts and recovered sales in the stores, and decide the scale-up on those numbers. The platform costs ~US\$ 440 a month at list price.

### WEEK 0 · SPONSOR

Pick the 4 pilot stores and the 4 control stores.

### WEEK 1 · LOJABR IT + DATABRICKS

Open the sales, stock and purchase-order feeds, and wire the approval into the ERP.

### WEEK 4 · STEERING COMMITTEE

Decide the scale-up with the measured result in hand.

# What sits behind each number

- **Lakeflow Declarative Pipelines:** Auto Loader on 7 raw feeds; medallion with 7 data-quality expectations, 0 failures in the last run.
- **Model:** gradient boosting on *point-in-time* features (the open order with its **expected** arrival, not the actual one). Trained on 153 k cases and tested on 49.7 k from later dates; AUC 0.85; retrained and registered in Unity Catalog on every run.
- **Agent:** tool-calling on the Foundation Model API; it queries 3 tools, and the system rejects a decision made before them.
- **Lakebase:** serving schema published atomically and app schema with approvals, rejections and the Genie log; reads in ~3 ms.
- **Genie:** entity matching, metric definitions and 5 certified SQL queries.
- **App:** React + TypeScript / FastAPI; service principal with least privilege; sign-in through the workspace SSO.
- **Evidence as text:** executed notebooks with outputs, data-quality results, Genie and app responses, and the cost queries in evidence/.

# What it costs to run, and how it is operated

| COMPONENT                 | USAGE ON 2026-09-29 | US\$/DAY |
|---------------------------|---------------------|----------|
| Serverless job (4 tasks)  | 0.35 DBU            | 0.16     |
| Lakeflow pipeline         | 0.39 DBU            | 0.17     |
| Agent (LLM, 24 calls)     | 2.72 DBU            | 0.19     |
| Lakebase (scales to zero) | 5.11 DBU            | 2.66     |
| App (always on)           | 12 DBU              | 11.40    |
| Total                     | ~US\$ 440 / 30 days | 14.58    |

Source: `system.billing.usage × list_prices`, on a day with only the scheduled run. The app is 78% of the cost and can be stopped outside business hours. Contract discounts lower these figures.

- **Latency:** daily batch at 06:00 with data through yesterday. A 2.9-day warning fits a daily batch, and Auto Loader can run through the day if needed.
- **Retraining and feedback:** the model is retrained every day. Approvals and rejections are stored with their author and become an evaluation signal in the pilot.
- **Scale:** compute is serverless. In the pilot we measure time and cost with the full assortment of the 4 stores.
- **Security:** role-based permissions in Unity Catalog, least-privilege access for the app, and every decision auditable.